

Database Schema

Database: PostgreSQL
ORM: Prisma ([backend/prisma/schema.prisma](#))
Last updated: June 2026

Overview

TracSOpro uses a multi-tenant schema centered on **SecurityCompany**. Users belong to the platform with a **Role** enum; guards, clients, and admins are linked to companies through join tables. All migrations live in [backend/prisma/migrations/](#).

SecurityCompany

├─ CompanyUser

├─ CompanyGuard

├─ CompanyClient

├─ CompanySite

├─ Invitation

├─ Subscription

└─ BillingRecord

→ User (ADMIN)

→ Guard → User (GUARD)

→ Client → User (CLIENT)

→ Site

Site → Shift → ShiftBreak, ShiftIncident, ShiftReport, ShiftCheckpoint

Guard → TrackingRecord, GeofenceEvent, IncidentReport

Enums

Enum	Values
Role	GUARD, ADMIN, CLIENT, SUPER_ADMIN
AccountType	INDIVIDUAL, COMPANY
GuardStatus	ACTIVE, ON_DUTY, OFF_DUTY, ON_LEAVE, SUSPENDED, TERMINATED
ShiftStatus	SCHEDULED, IN_PROGRESS, ON_BREAK, COMPLETED, CANCELLED, NO_SHOW, EARLY_END
IncidentType	SECURITY_BREACH, THEFT, VANDALISM, SUSPICIOUS_ACTIVITY, MEDICAL_EMERGENCY, FIRE, NATURAL_DISASTER, EQUIPMENT_FAILURE, OTHER
IncidentSeverity	LOW, MEDIUM, HIGH, CRITICAL
IncidentStatus	REPORTED, INVESTIGATING, RESOLVED, CLOSED, ESCALATED
SubscriptionPlan	BASIC, PROFESSIONAL, ENTERPRISE, CUSTOM
SubscriptionStatus	TRIAL, ACTIVE, SUSPENDED, CANCELLED, EXPIRED
GeofenceEventType	ENTER, EXIT

Enum	Values
BreakType	REGULAR, LUNCH, EMERGENCY, UNAUTHORIZED
NotificationType	SHIFT_REMINDER, INCIDENT_ALERT, MESSAGE, SYSTEM, EMERGENCY
ConversationType	DIRECT, GROUP, TEAM
SupportAudience	COMPANY, PLATFORM
SupportStatus	OPEN, IN_PROGRESS, RESOLVED, CLOSED

Core identity

User

Central authentication record for all roles.

Field	Type	Notes
id	UUID	Primary key
email	String	Unique
password	String	bcrypt hash
firstName, lastName	String	Display name
role	Role	GUARD, ADMIN, CLIENT, SUPER_ADMIN
accountType	AccountType?	INDIVIDUAL or COMPANY
isEmailVerified	Boolean	OTP verification gate
isActive	Boolean	Soft disable

Relations: one-to-one with [Guard](#) or [Client](#); many [RefreshToken](#), [DeviceToken](#), [Notification](#).

Guard

Field officer profile linked to [User](#).

Field	Type	Notes
employeeId	String	Unique per guard
status	GuardStatus	ACTIVE, ON_DUTY, etc.
certificationUrls	String[]	Document URLs

Relations: [Shift\[\]](#), [TrackingRecord\[\]](#), [IncidentReport\[\]](#), [CompanyGuard\[\]](#).

Client

Client organization profile linked to [User](#).

Field	Type	Notes
accountType	AccountType	INDIVIDUAL or COMPANY
companyName	String?	For company accounts

Relations: `Site[]`, `Shift[]`, `CompanyClient[]`.

Operations

Site

Physical location managed by a client.

Field	Type	Notes
clientId	UUID	Owner client
latitude, longitude	Float?	Map coordinates
radiusMeters	Int	Geofence radius (default 100)
isActive	Boolean	Soft disable

Shift

Work assignment linking guard, site, and client.

Field	Type	Notes
guardId	UUID?	Nullable until assigned
siteId, clientId	UUID?	Site and client refs
scheduledStartTime, scheduledEndTime	DateTime	Planned window
actualStartTime, actualEndTime	DateTime?	Check-in/out times
status	ShiftStatus	Lifecycle state
checkInLocation, checkOutLocation	Json?	GPS snapshot

Lifecycle: `SCHEDULED` → `IN_PROGRESS` → `(ON_BREAK)` → `COMPLETED`

ShiftBreak

Break periods during an active shift.

Field	Type	Notes
shiftId	UUID	Parent shift
startTime, endTime	DateTime?	Break window
breakType	BreakType	REGULAR, LUNCH, etc.

TrackingRecord

GPS history for a guard.

Field	Type	Notes
guardId	UUID	Guard reference
latitude, longitude	Float	Coordinates
accuracy	Float?	GPS accuracy (meters)
batteryLevel	Int?	Device battery
timestamp	DateTime	Point-in-time

GeofenceEvent

Enter/exit events for site geofences.

Field	Type	Notes
guardId	UUID	Guard reference
geofenceId	String	Site or geofence identifier
eventType	GeofenceEventType	ENTER or EXIT

Incidents and reports

Incident (legacy)

Location-based incident with evidence attachments.

Field	Type	Notes
reportedBy	UUID	User ID
locationId	UUID	Location reference
type, severity, status	Enums	Classification
evidence	Evidence[]	Photos, videos, documents

IncidentReport

Guard-submitted reports with media (primary mobile flow).

Field	Type	Notes
guardId	UUID	Submitting guard
reportType	String	Category
status	String	SUBMITTED, etc.

Field	Type	Notes
statusHistory	Json?	Status change log
media	IncidentReportMedia[]	Images/videos

ShiftIncident

Incident tied to an active shift.

Field	Type	Notes
shiftId	UUID	Parent shift
incidentType, severity	Enums	Classification
attachments	Json?	File URL array

ShiftReport

Text reports submitted by guards for a shift.

Field	Type	Notes
shiftId, guardId	UUID	References
reportType	ReportTypeEnum	SHIFT, INCIDENT, EMERGENCY
content	String	Report body

Communication

Conversation / Message

In-app chat with participant tracking.

Model	Key fields
Conversation	id, type (DIRECT/GROUP/TEAM), securityCompanyId, lastMessageAt
ConversationParticipant	conversationId, userId, role, lastReadAt
Message	senderId, conversationId, content, messageType, isRead

Notification

In-app notification inbox.

Field	Type	Notes
userId	UUID	Recipient
type	NotificationType	SHIFT_REMINDER, EMERGENCY, etc.
data	String?	JSON payload for deep links

Field	Type	Notes
isRead	Boolean	Read state

DeviceToken

FCM push registration per device.

Field	Type	Notes
userId	UUID	Owner
token	String	FCM token
platform	String	ios or android
isActive	Boolean	Token validity

PushNotificationRetry

Queue for failed push deliveries with exponential retry.

Multi-tenancy

SecurityCompany

Tenant root for each security company on the platform.

Field	Type	Notes
name, email	String	Company identity
subscriptionPlan	SubscriptionPlan	BASIC, PROFESSIONAL, etc.
subscriptionStatus	SubscriptionStatus	TRIAL, ACTIVE, etc.
maxGuards, maxClients, maxSites	Int	Plan limits
isActive	Boolean	Tenant enable/disable

CompanyUser

Links admin users to a company.

Field	Type	Notes
securityCompanyId, userId	UUID	Composite unique
role	CompanyRole	OWNER, ADMIN, MANAGER, etc.

CompanyGuard / CompanyClient / CompanySite

Join tables scoping guards, clients, and sites to a security company.

Invitation

Onboarding codes for guard and client registration.

Field	Type	Notes
invitationCode	String	Unique code
role	Role	GUARD or CLIENT
expiresAt	DateTime	Expiration
maxUses, currentUses	Int	Single or multi-use
isActive	Boolean	Revocable

Billing

Subscription

Stripe-linked subscription per company.

Field	Type	Notes
stripeSubscriptionId	String?	Stripe reference
plan, status	Enums	Plan and lifecycle
billingCycle	BillingCycle	MONTHLY, QUARTERLY, YEARLY

BillingRecord

Invoices and payment records.

Field	Type	Notes
type	BillingType	SUBSCRIPTION, OVERAGE, etc.
status	BillingStatus	PENDING, PAID, OVERDUE
stripeInvoiceId	String?	Stripe reference

StripeWebhookEvent

Idempotency log for processed Stripe webhooks.

Platform

PlatformSettings

Key-value settings scoped globally or per company.

PlatformAnalytics

Time-series metrics (active guards, revenue, etc.).

SystemAuditLog

Audit trail for super-admin actions (impersonation, config changes).

Auth tokens

RefreshToken

Rotating refresh tokens with revocation support.

Field	Type	Notes
jti	String	Unique token ID
revokedAt	DateTime?	Revocation timestamp
expiresAt	DateTime	Expiration

Support

SupportTicket / SupportTicketReply

Help desk tickets with company or platform audience.

Field	Type	Notes
audience	SupportAudience	COMPANY or PLATFORM
securityCompanyId	String?	Tenant scope
conversationId	String?	Linked chat thread

Checkpoint (partial)

QR checkpoint scanning schema exists; mobile UI is limited.

Model	Purpose
Checkpoint	QR code tied to a Location
ShiftCheckpoint	Scan record during a shift

Indexes and tenancy

- Most tenant-scoped queries filter by `securityCompanyId` via join tables.
- `User.email` and `User.role` are indexed.
- `TrackingRecord.timestamp` and `Shift.scheduledStartTime` are indexed for time-range queries.
- Tenant isolation is enforced in services and tested in `backend/tests/tenantIsolation.test.ts`.

Migrations

```
cd backend
npx prisma migrate dev      # Development: create migration
npx prisma migrate deploy   # Production: apply migrations
```



```
npx prisma generate      # Regenerate Prisma client
npx prisma db push       # Dev only: push schema without migration
```

Related documents

- [SYSTEM_ARCHITECTURE.md](#) — domain flows
- [API_REFERENCE.md](#) — REST endpoints
- [DEPLOYMENT_GUIDE.md](#) — production migration workflow